

- 4 The matrices **A**, **B** and **C** are given by

$$\mathbf{A} = \begin{pmatrix} 2 & k & k \\ 5 & -1 & 3 \\ 1 & 0 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ and } \mathbf{C} = \begin{pmatrix} 0 & 1 & 1 \\ -1 & 2 & 0 \end{pmatrix},$$

where k is a real constant.

- (a) Find **CAB**.

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- (b) Given that **A** is singular, find the value of k .

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- (c) Using the value of k from part (b), find the equations of the invariant lines, through the origin, of the transformation in the x - y plane represented by **CAB**. [5]

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